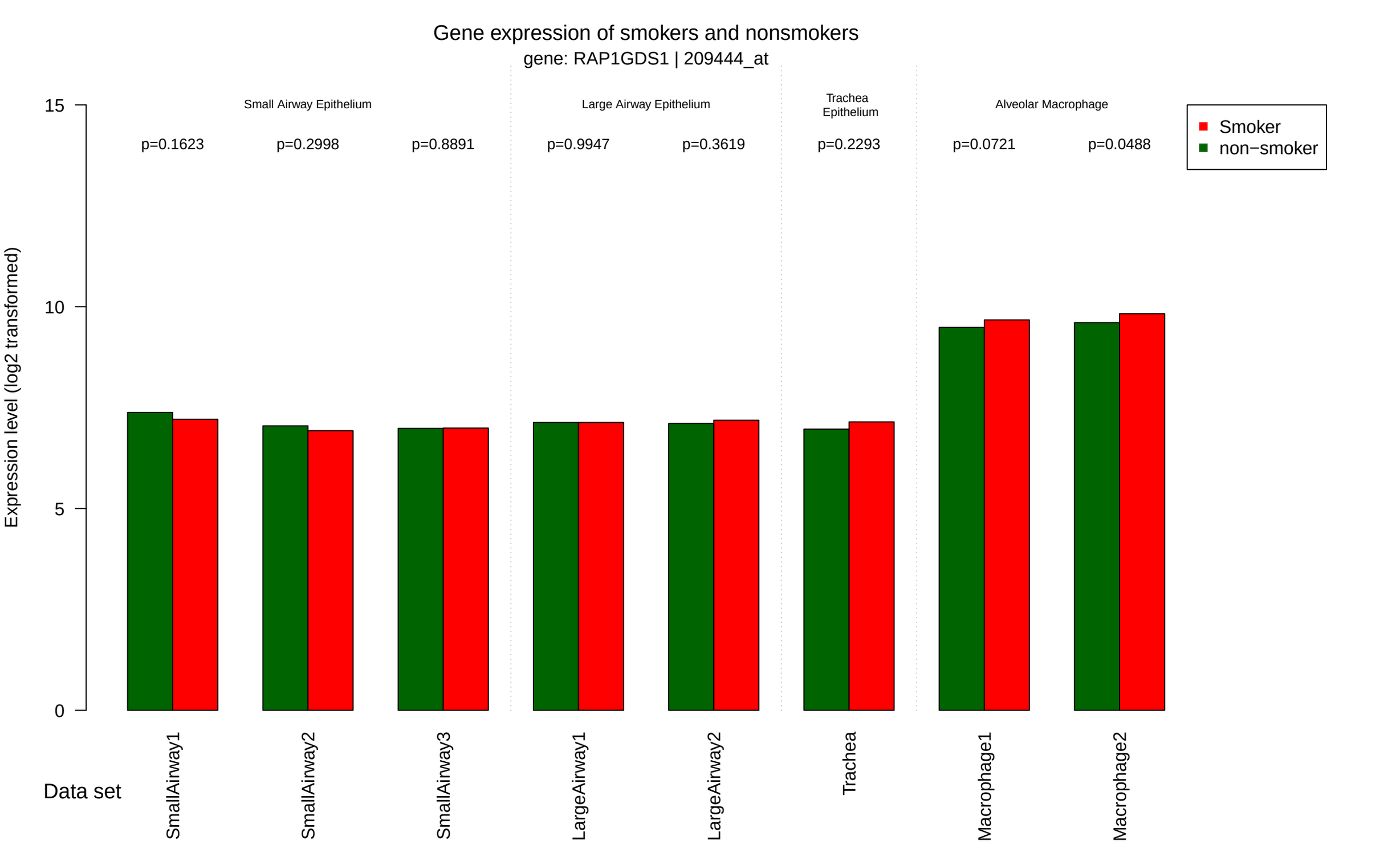
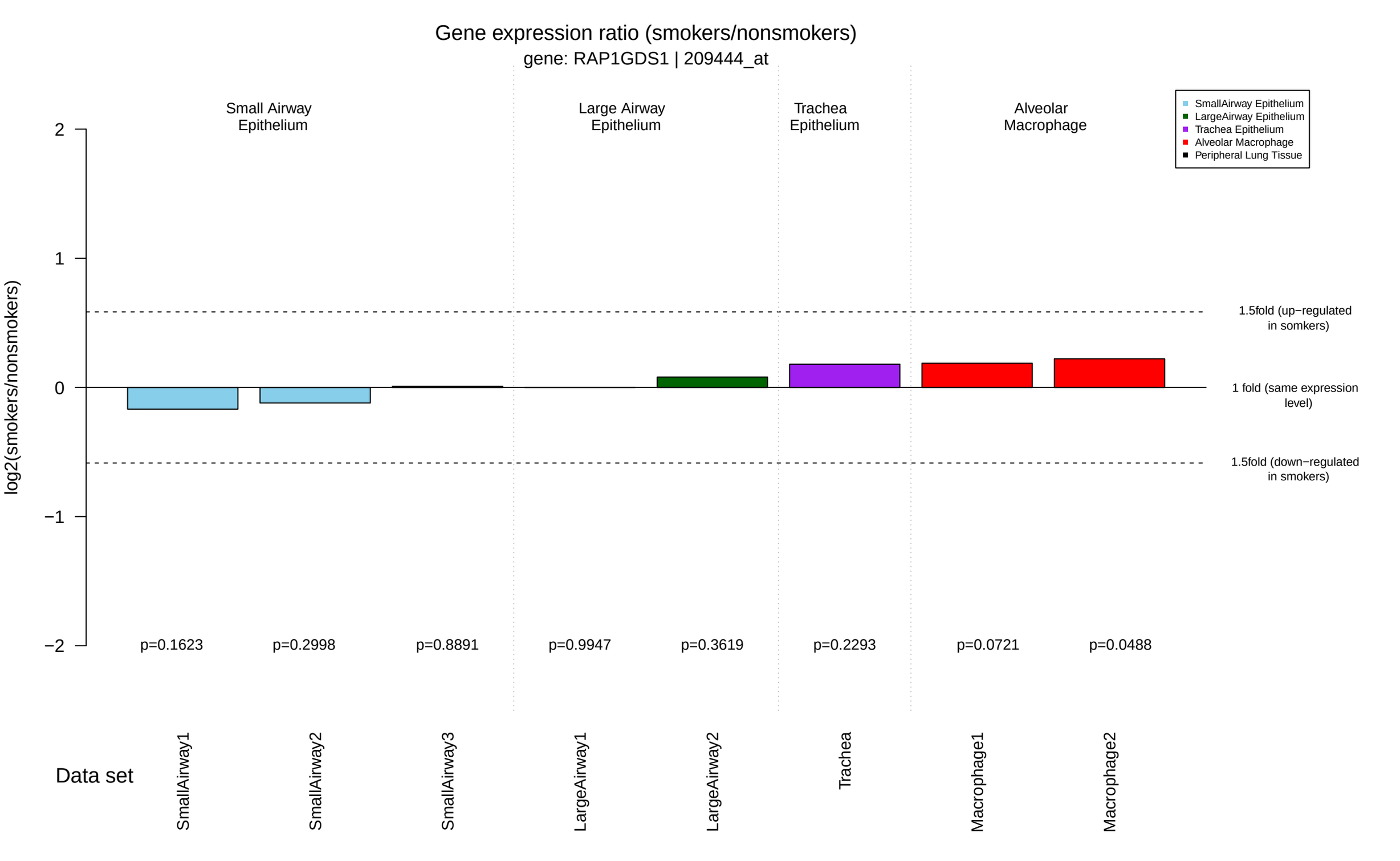
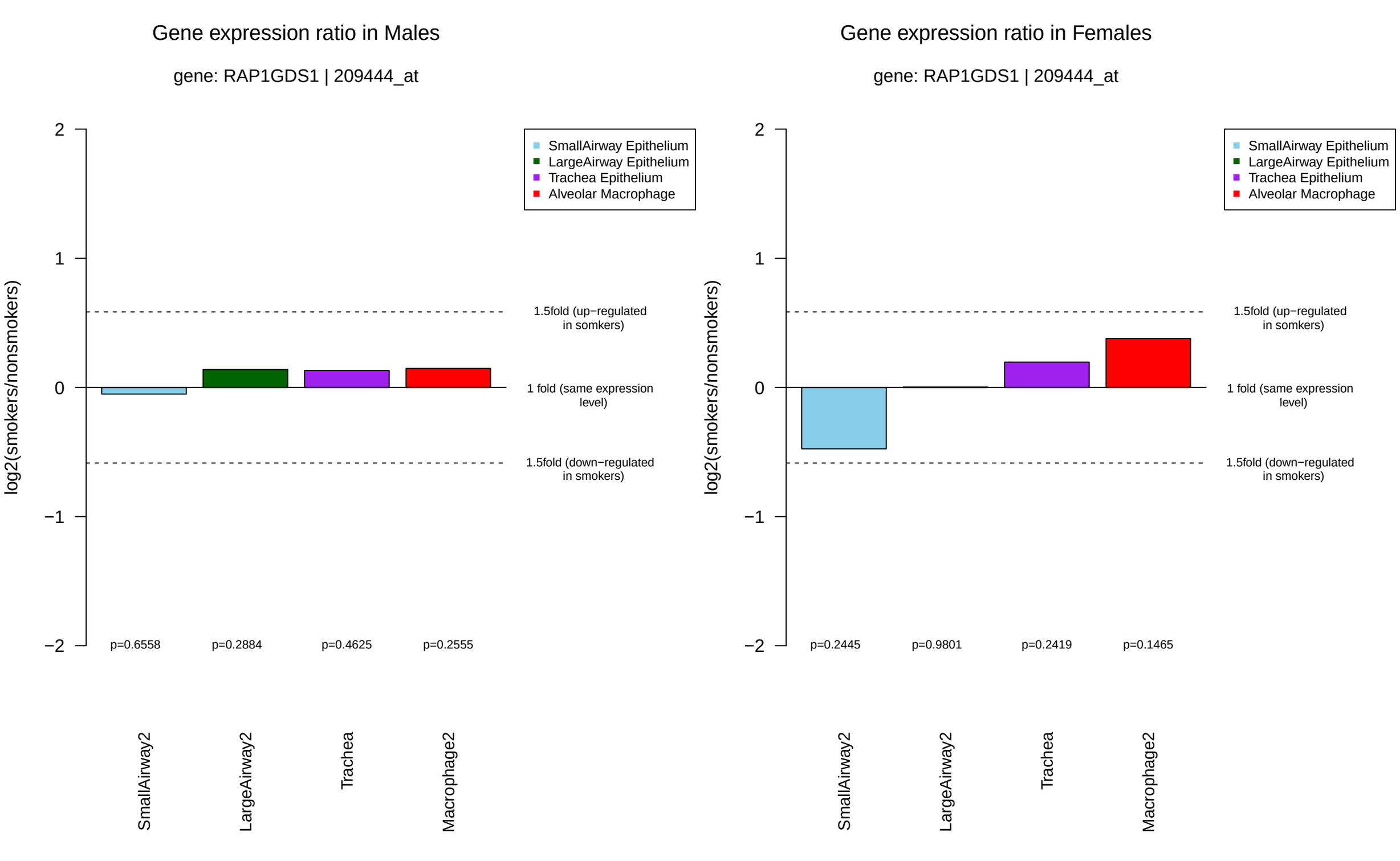
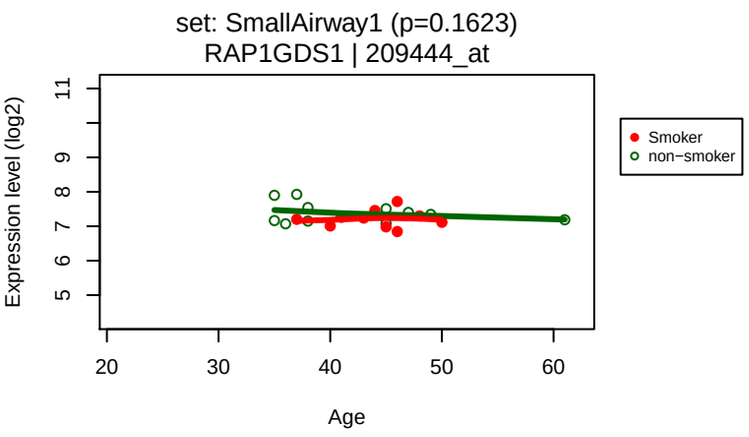
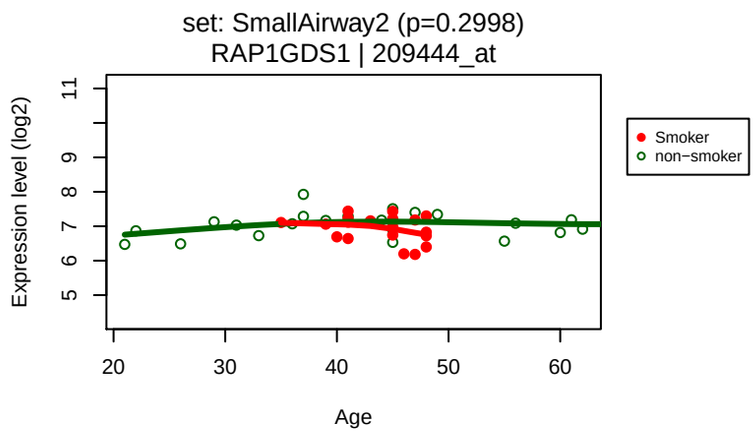
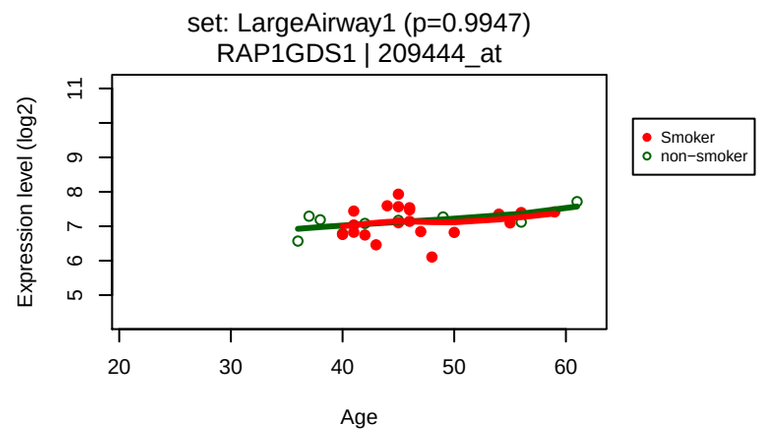

Gene expression of smokers and nonsmokers
gene: RAP1GDS1 | 209444_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1623
p=0.2998
p=0.8891
p=0.9947
p=0.3619
p=0.2293
p=0.0721
p=0.0488
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

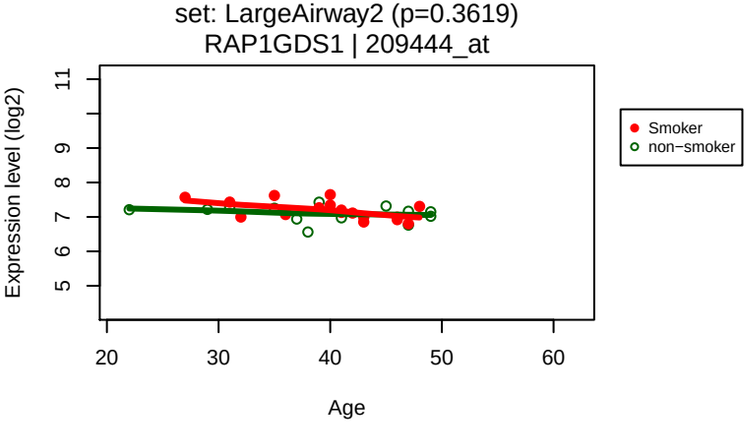

Gene expression ratio (smokers/nonsmokers)
gene: RAP1GDS1 | 209444_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1623
p=0.2998
p=0.8891
p=0.9947
p=0.3619
p=0.2293
p=0.0721
p=0.0488
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

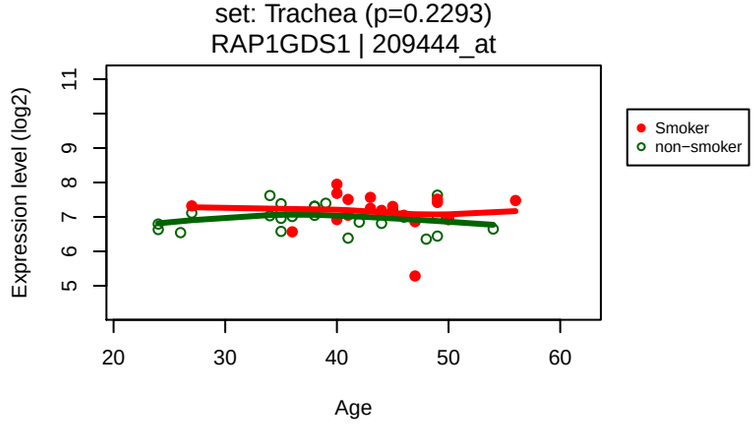

Gene expression ratio in Males
gene: RAP1GDS1 | 209444_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6558
p=0.2884
p=0.4625
p=0.2555
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAP1GDS1 | 209444_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2445
p=0.9801
p=0.2419
p=0.1465
SmallAirway2
LargeAirway2
Trachea
Macrophage2

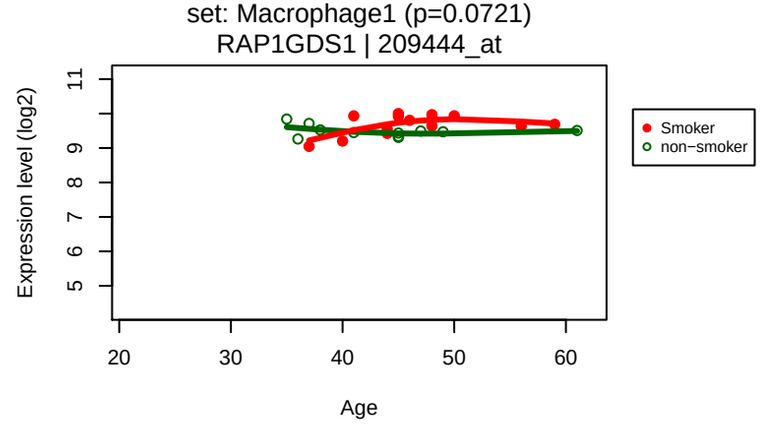

set: SmallAirway1 (p=0.1623)
RAP1GDS1 | 209444_at
Expression level (log2)
Age
Smoker
non−smoker

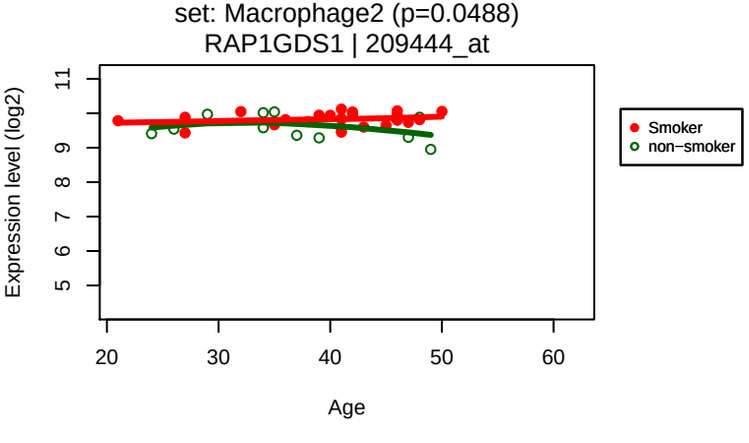

set: SmallAirway2 (p=0.2998)
RAP1GDS1 | 209444_at
Expression level (log2)
Age
Smoker
non−smoker

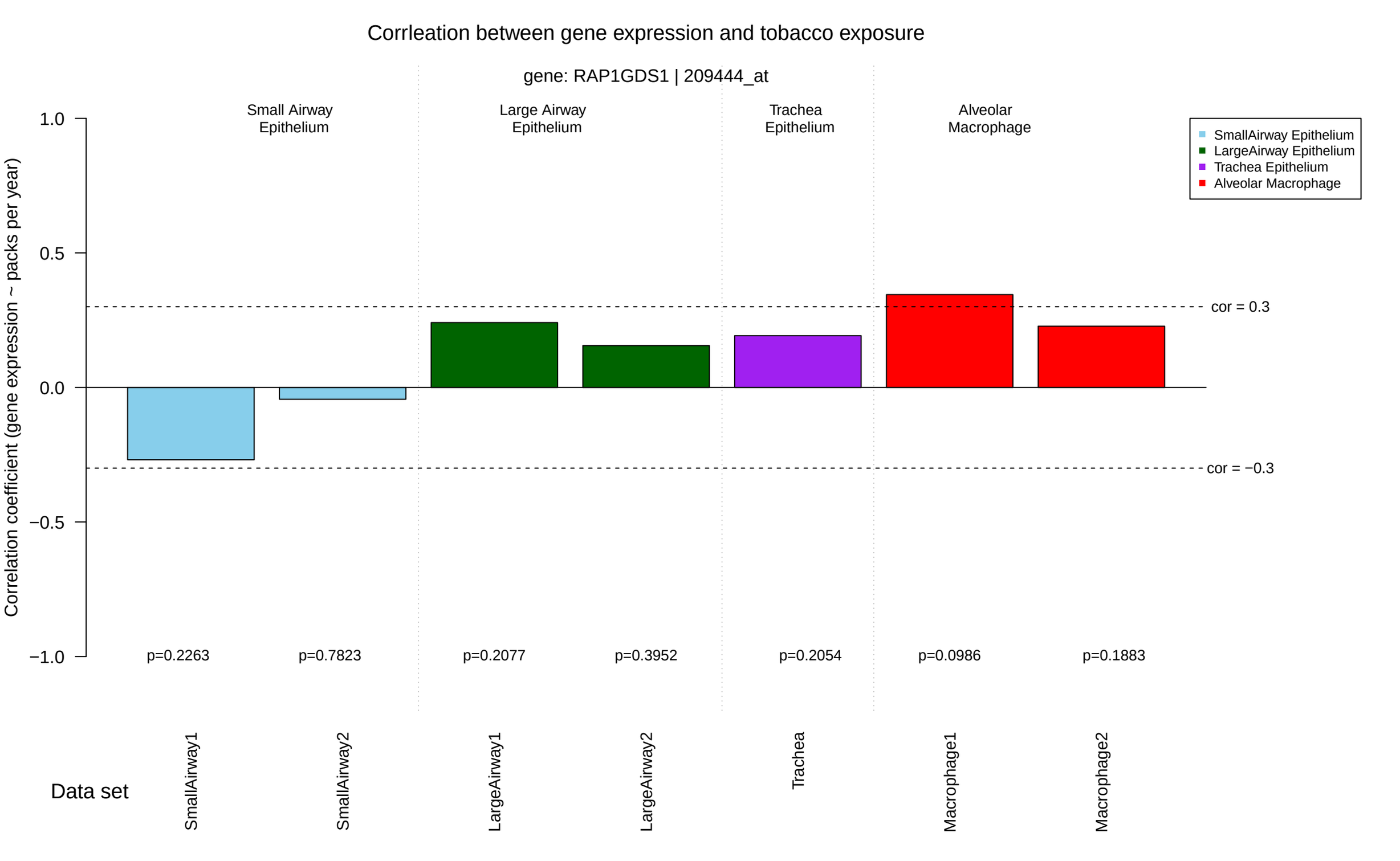

set: LargeAirway1 (p=0.9947)
RAP1GDS1 | 209444_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3619)
RAP1GDS1 | 209444_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2293)
RAP1GDS1 | 209444_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0721)
RAP1GDS1 | 209444_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0488)
RAP1GDS1 | 209444_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAP1GDS1 | 209444_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2263
p=0.7823
p=0.2077
p=0.3952
p=0.2054
p=0.0986
p=0.1883
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2